

Task 1: Introduction

To defend against attacks, a **SOC** team relies on various security solutions, such as **SIEM**, **EDR**, firewalls, and threat intelligence platforms. They also communicate with **IT** and management teams as part of their processes.

However, as threats grow more complex and advanced, **SOC** teams face challenges like alert fatigue, manual processes, too many disconnected tools, and difficulties in communication across teams.

In this room, we will explore how the Security Orchestration, Automation, and Response (**SOAR**) tool overcomes these challenges for a **SOC** team.

SOAR = Smarter
SOC, Stronger
Defense!



Learning Objectives



- ① Understand the traditional **SOC** and its challenges
- ② Explore how **SOAR** overcomes these challenges
- ③ Learn **SOAR** Playbooks
- ④ Practically walk through a threat intelligence workflow



SOAR helps
SOC teams
work smarter,
not harder!

Room Prerequisites



A look at the following rooms would be helpful before starting this room:

- SOC Fundamentals
- Junior Security Analyst Intro
- Intro to Detection Engineering

Key Takeaway



SOAR empowers SOC teams by automating repetitive tasks, integrating tools, and enabling faster, smarter responses to modern threats.

Task 2: Traditional SOC and Challenges

★ How Traditional SOC's Work

Organisations have a Security Operations Center (SOC) as a centralised location for monitoring and protecting their digital assets. SOC's have evolved over time, with every generation adding new technology. The main advantage of an organisation having a SOC is to enhance its security incident handling through continuous monitoring and analysis. This is achievable by implementing the right amount of people, processes, and technologies to support the SOC's capabilities and business goals.

Some key capabilities of a SOC include the following:

- ① **Monitoring and Detection:** Continuously scanning and flagging suspicious activities within a network environment. It leads to awareness of emerging threats and how to prevent them in their early stages. The monitoring is mainly done through the SIEM.

Example: Detecting numerous failed login attempts on a critical workstation, or detecting a login from an unknown location, etc.

- ② **Recovery and Remediation:** SOC teams operate as first responders when cyber threats are identified. They perform operations such as isolating or shutting down infected endpoints, removing malware, and stopping malicious processes. During this process, they often utilize other security solutions like EDR, firewalls, IAM, etc.

Example: Isolating an endpoint through EDR, blocking an IP on the firewall, disabling a user on the IAM, etc.

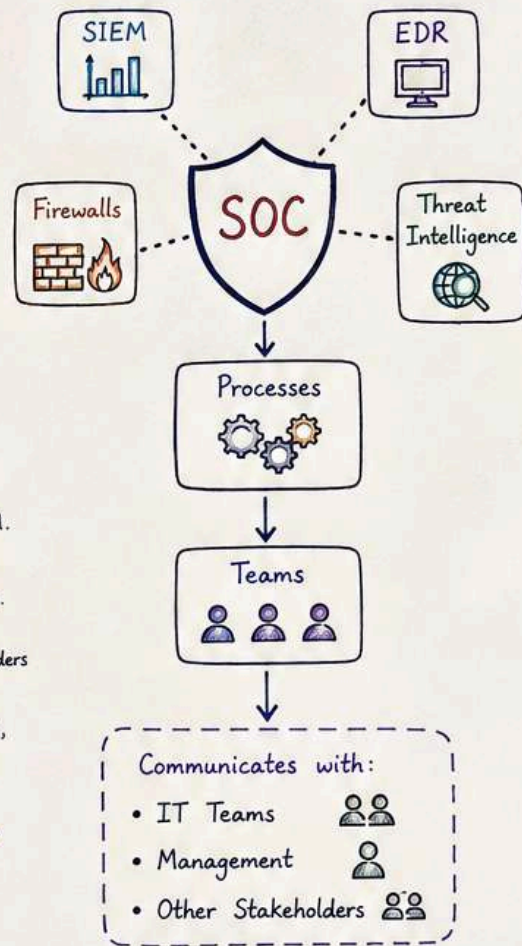
- ③ **Threat Intelligence:** Monitoring environments continuously requires a constant flow of threat intelligence. This ensures that SOC teams have continuous and the latest feeds of threat data, such as IP addresses, hashes, domains, and other indicators.

Example: Blocking a malicious domain flagged by the threat intelligence feeds.

- ④ **Communication:** SOC teams not only detect and respond to threats but also coordinate with IT teams and management to effectively communicate the threats and ensure that the incidents are addressed.

Example: Generating a ticket for the IT team to verify a recently deployed patch.

How a Traditional SOC Operates



SOCs leverage multiple tools and collaborate with various teams to detect, respond, and recover from cyber threats.

★ Challenges Faced by SOC's

	1) Alert Fatigue	Using numerous security tools triggers a large number of alerts within an SOC. Many of these alerts are false positives or insufficient for an investigation, leaving analysts overwhelmed and unable to address any serious security events.
	2) Too many Disconnected Tools	Security tools are often deployed without integration within an organisation. Security teams are tasked with navigating through firewall logs and rules, which are handled independently from endpoint security logs. This also leads to an overload of tools.
	3) Manual Processes	SOC investigation procedures are often not documented, leading to inefficient means of addressing threats. Most rely on established tribal knowledge built by experienced analysts, and the processes are never documented. This results in slowing down the investigation and increasing response times.
	4) Talent Shortage	SOC teams find recruiting and expanding their talent pool difficult to address the growing security landscape and sophisticated threats. Combining this with the alert overload teams face, security analysts become more overwhelmed with the responsibilities they have to undertake, resulting in less efficient work and extended incident response times that allow adversaries to wreak havoc within an organisation.



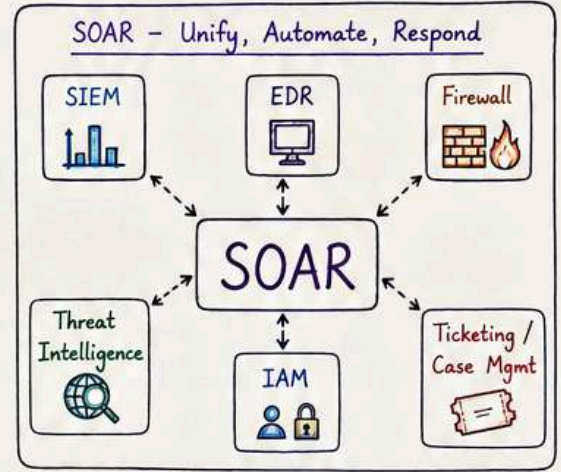
Bottom line: More tools and alerts $\neq$ More Security.
It's about smarter processes, automation, and integration!

Task 3: Overcoming SOC Challenges with SOAR

In the previous task, we saw some challenges that Security Operations Centers (SOC) faced. In this task, we will learn about a tool that can overcome these challenges for a SOC team. This tool is called Security Orchestration, Automation, and Response (SOAR). Let's discuss what this tool is and how it overcomes these challenges.

What is SOAR?

Security Orchestration, Automation, and Response (SOAR) is a tool that unifies all the security tools used in a SOC. With SOAR, SOC analysts do not need to switch between SIEM, EDR, Firewall, and other security tools for their investigations. They can operate all these tools within a single SOAR interface. Along with unifying the security tools, it also provides ticketing and case management features to the analysts, through which they can document, track, and resolve their incidents in a structured way.



Benefits of SOAR

- ✓ Reduces alert fatigue
- ✓ Automates repetitive tasks
- ✓ Connects all tools in one place
- ✓ Speeds up incident response
- ✓ Structured case management

The core strength of a SOAR tool comes from the following three main capabilities:

① Orchestration

It connects different tools from various vendors within the unified SOAR interface. It defines workflows for investigating various types of alerts, known as Playbooks. These playbooks are predefined steps that tell the SOAR how to investigate an alert.

Example: VPN Brute Force Alert Playbook

- ① Received alert from SIEM
- ② Query SIEM to check if the User normally uses the IP
- ③ Check TI platforms for the IP's reputation
- ④ Query SIEM for any successful logins
- ⑤ Escalate to containment actions

These playbooks are dynamic and usually contain different paths. The result of each step determines the next action.



Without Orchestration (Traditional)

- Open SIEM
 - Switch to TI Platform
 - Switch to IAM
 - Switch to Ticketing System
 - (repeat)
- Time Consuming!

② Automation

The orchestration (playbooks) can be automated. Automation means no more manual clicks needed from SOC analysts. SOAR will itself follow the playbooks.

Example: VPN Brute Force Alert Playbook with Automation

- ⚙️ ① SOAR receives the alert from SIEM
- ⚙️ ② It automatically queries the SIEM for the user's historical logins
- ⚙️ ③ It automatically verifies the IP's reputation through TI platforms
- ⚙️ ④ If the IP is malicious, it automatically disables the user from the IAM
- ⚙️ ⑤ Lastly, it automatically opens a ticket in the ticketing system with all the details to initiate an investigation

Saves a tremendous amount of time for SOC analysts! They can handle hundreds of alerts without burning out. 😊

Manual vs Automated

Manual (Traditional)

- Lots of manual steps 😞
- Time consuming
- Prone to human error

Automated (SOAR)

- Follows playbooks automatically 😊
- Fast and consistent
- More accurate
- Frees up analyst time

③ Response

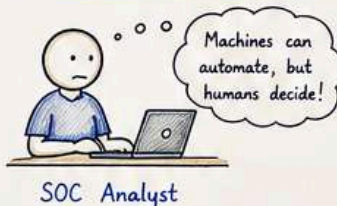
SOAR gives the ability to take actions using different tools from one unified interface. It also automates the response.

Example Response Actions (via SOAR Playbook)



SOAR can follow the playbook of VPN Brute force and take all these actions automatically.

Do We Still Need SOC Analysts?



While a SOAR tool can automate the majority of repetitive tasks, it does not replace SOC analysts. Complex investigations still require an SOC analyst. SOAR cannot give a judgment call at some critical points, but an analyst can. A SOC analyst understands the threats in the broader business context. The playbooks for different types of alerts are also made by the SOC analysts.

Bottom Line

SOAR would ease the burden of SOC by automating repetitive tasks and organizing everything in a simplified structure, but we still need SOC analysts. ★

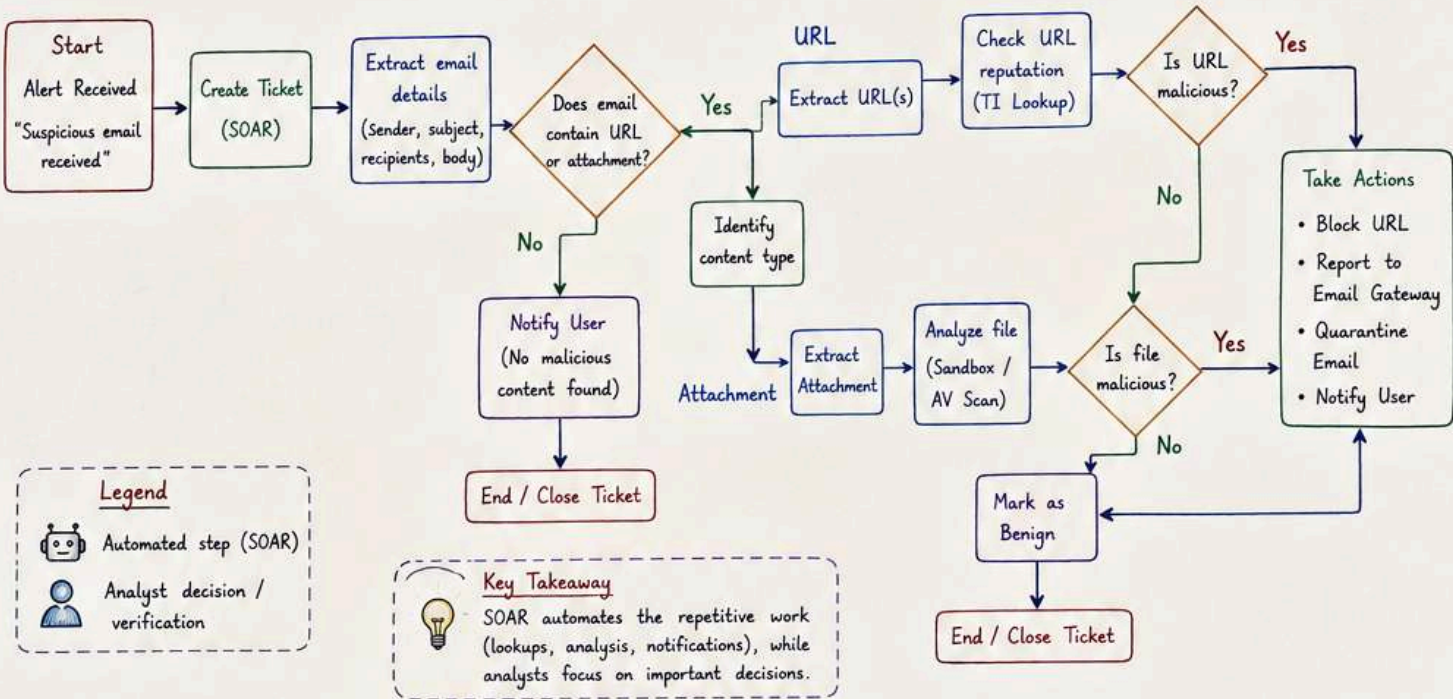
Task 4: Building SOAR Playbooks

Playbooks are predefined workflows that tell SOAR what actions to take during a specific investigation.

In the previous task, we studied how SOAR Playbooks are predefined workflows that tell the SOAR tool what actions to take during a specific investigation. SOC analysts make playbooks for a general category of recurring alerts. Let's have a look at two different playbooks, one for Phishing and one for CVE Patching.

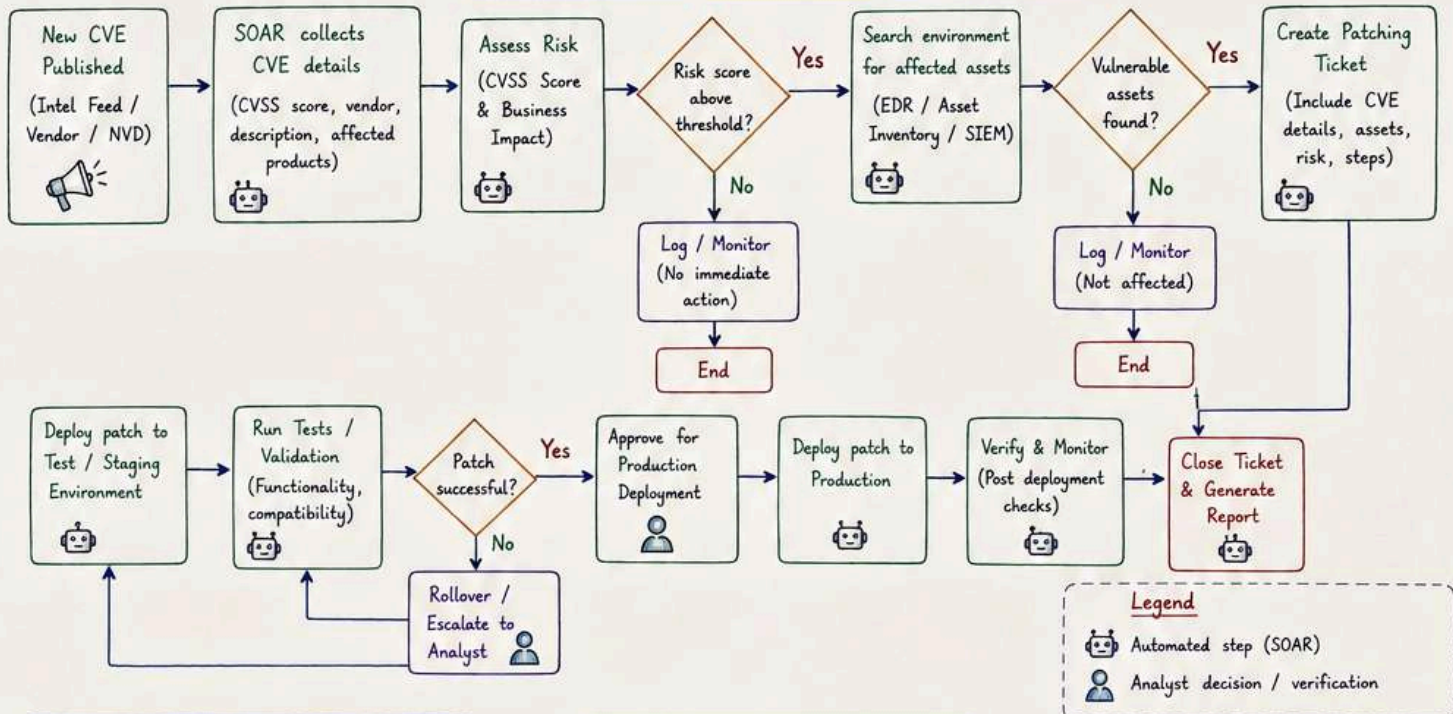
1) Phishing Playbook

Goal: Detect, analyze and respond to phishing emails to protect users and the organization.



2) CVE Patching Playbook

Goal: Identify new CVEs, assess risk, create patching ticket, test the patch and deploy to production.



★ Important Note

Most steps in a playbook are automated, but analyst input is essential at key decision points.

SOAR Playbooks help SOC teams to:

- ✓ Save time and resources
- ✓ Reduce manual errors
- ✓ Ensure consistent and faster response
- ✓ Improve overall security posture



Work Smarter,
Not Harder!